AI-Powered School Attendance Management

Graduation Capstone Project Proposal - Education

Founders:

- **Rana Mohamed** (Leader):
- **Nouran Hesham**: MA Candidate in International Political Economy, Researcher
- **Yara Osama**:
- **Asmaa Maher**:

Instructor: Eng. Baraa Abu Sallout

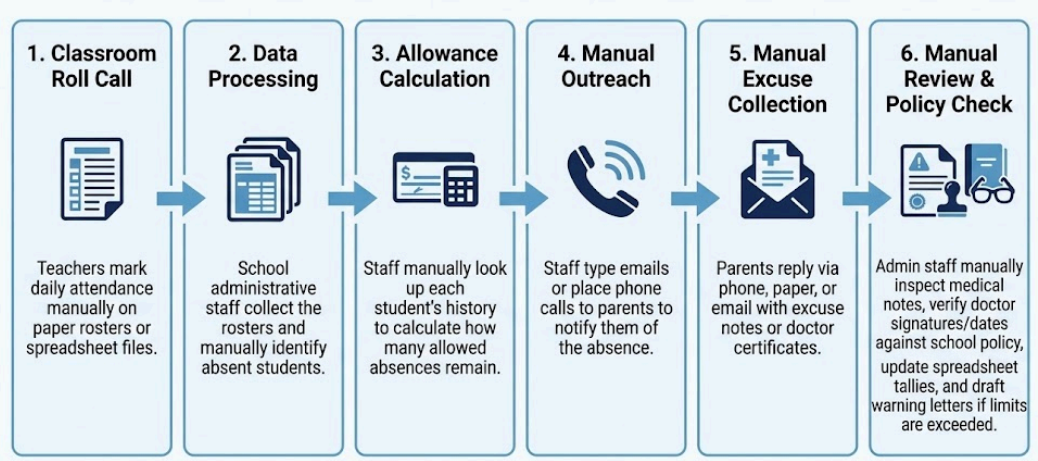
1. Project Title

> ***AIAttend Pro*** - An AI-Powered Automated Attendance System
> *A Smart Student Attendance & Automated Early Warning Workflow*

2. Business Area and Target User

Business Unit:	**Education Operations & Student Administration**
Primary End-User Role:	**School Administrator** *who tracks daily attendance, processes medical excuses, and manages rosters*
Secondary Beneficiary	**Parents / Guardians** *who receive real-time notifications and remaining absence allowance alerts*

3. Current Process (Manual/ Traditional)



Step #	Task	Manual Action & Current Owner	Output
Step 1	**Classroom Roll Call**	Teachers mark daily attendance manually on paper rosters or spreadsheet files	Marked check of attendance
Step 2	**Data Processing**	School administrative staff collect the rosters and manually identify absent students	
Step 3	**Allowance Calculation**	Staff manually look up each student's history to calculate how many allowed absences remain	Calculated used days vs. remaining days
Step 4	**Manual Outreach**	Staff type emails or place phone calls to parents to notify them of the absence	Calls, emails, and notes on responses
Step 5	**Manual Excuse Collection**	Parents reply via phone, paper, or email with excuse notes/doctor certificates.	Medical notes filed in student folders
Step 6	**Manual Review & Policy Check**	Admin staff manually inspect medical notes, verify doctor signatures/dates against school policy, update spreadsheet, and draft warning letters if limits are exceeded.	

4. Main Pain Point and Current Impact

- **Extreme Time Waste:** Staff spend hours daily manually collecting paper rosters, calculating remaining allowances, and making individual outreach calls.
- **Delayed Interventions:** Slow manual processing delays notifications, causing warning letters to arrive too late and putting students at risk of failing or dropping out.
- **High Compliance & Error Risks:** Manual record-keeping leads to lost medical notes, math errors, and inconsistent policy enforcement.
- **Parent Disputes:** The lack of immediate, automated communication logs creates friction with parents who claim they were never informed.

5. Exact Task to Automate First

To build an end-to-end, automated attendance and excuse management system in n8n that replaces manual daily roll calls, allowance calculations, parent notifications, and excuse processing through a two-stage workflow:

- **Daily Check-In & Threshold Alerting (Phase A):** Automatically fetch daily classroom attendance, cross-reference student history in Google Sheets/database to calculate remaining allowed days, and dispatch instant personalized WhatsApp/Email alerts to parents with auto-calculated risk thresholds and escalation triggers.
- **Excuse Ingestion, Data Aggregation & Policy Verification (Phase B):** Automatically receive incoming parent replies (text or attachments), route attachments through OCR, aggregate

extracted document data alongside student history and RAG-retrieved school handbook policy rules, and run a single LLM evaluation to generate a structured decision.

6. Trigger, Input Data, and Expected Output

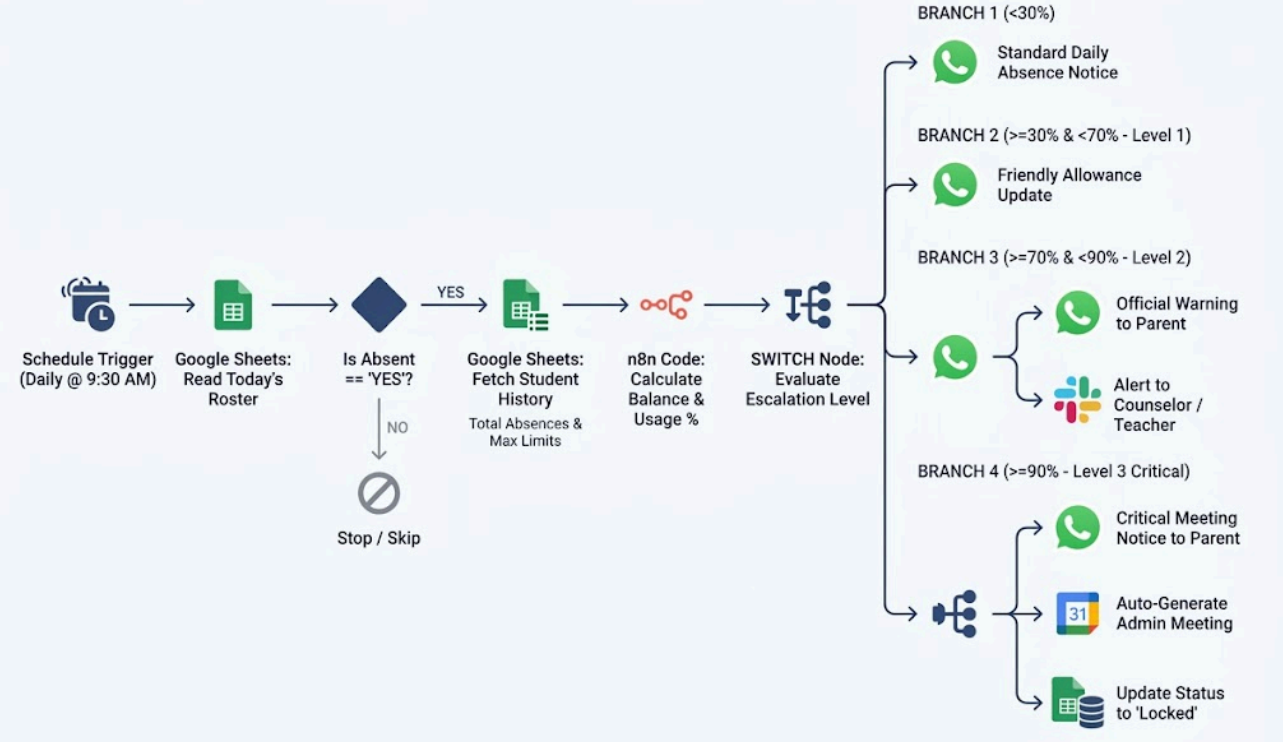
Stage	Workflow 1: Daily Check-In & Threshold Alerting	Workflow 2: Absence Justification, Medical Note Verification & Escalation
Trigger Event:	Scheduled Trigger (by the end of each class day)	Webhook Trigger: Whenever a parent replies to an outreach email or WhatsApp message.
Input Data Format:	1. **Daily Attendance Roster:** Today's check-in/roll-call records *(Student ID, Class, Status: Present/Absent/Late)* from Google Sheets. 2. **Student Absence History:** Historical total absences, max allowed limit per term, and parent contact details (Phone/Email).	1. **Parent Message Payload:** Text reply and optional attachment binary/URL (image or PDF of medical certificate/note). 2. **Student Profile & History:** Current database record. 3. **RAG Policy Context:** Relevant clauses retrieved from the official Student Attendance Handbook vector index.
Expected Output:	**1. Calculated Metrics:** Updated remaining allowed days and absence usage percentage. **2. Outreach Messages**: Automated, personalized WhatsApp/Email notifications sent to parents of absent students stating remaining days and requesting excuse documents if applicable. **3. Escalations**: Automated alerts to academic counselors or calendar invitations generated for critical absence thresholds (e.g., 90%+ allowance used).	1. **HITL Approval Request**: Admin dashboard/Slack notification displaying document image, AI recommendation, and 1-click [Approve] or [Reject] buttons. 2. **Final Database & Notification Action**: - ***On Approval:*** Google Sheets updated to "Excused", remaining days balance recalculated, and confirmation sent to parent. - ***On Rejection:*** Record maintained as "Unexcused", rejection reasons logged, and clarification request sent to parent.

7. Workflow (Nodes / Simple Flow)

- **Workflow 1 (Daily Check-In & Threshold Alerting):**
  Daily roll call/kiosk check-in → recalculating remaining allowed absences → instant parent notifications (WhatsApp/SMS) with exact days left.
- **Workflow 2 (Absence Justification, Medical Note Verification & Escalation):**
  Parent submits a justification/medical note → validating the excuse → adjusting the absence counter → triggering warning levels or scheduling parent-admin meetings.

Workflow 1: Daily Attendance Check & Warning System

(Runs automatically every morning when attendance is marked)



Node 1: Schedule Trigger Node Fires automatically every school morning at a set time (e.g., 9:30 AM) after teachers finish roll call.

Node 2: Read Today's Roster Node: Queries the daily attendance roster table or Google Sheet and filters for rows where is_absent == "YES".

Node 3: Fetch Student History Node: Pulls historical record data for each missing student, including total recorded absences, maximum term allowance, and parent contact details.

Node 4: Code / Calculation Node: Executes straightforward math to compute metrics:

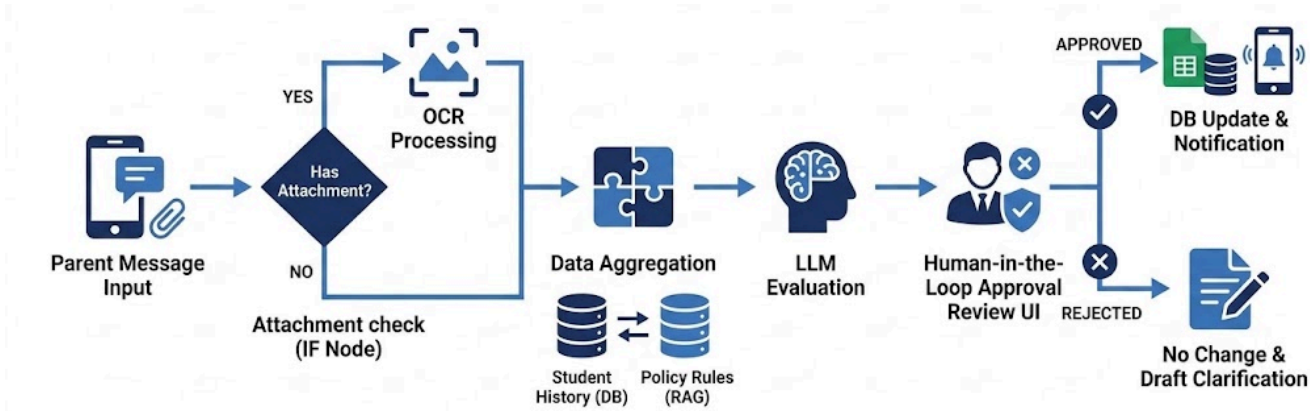
- $TotalAbsencesToday = PreviousAbsences + 1$
- $RemainingDaysLeft = MaxAllowed - TotalAbsences$
- $PercentageUsed = \left(\frac{TotalAbsences}{MaxAllowed}\right) \times 100$

Node 5: Switch / Router Node: Routes the message execution along 4 dynamic threshold branches:

- ***Branch 1*** (< 30% Used): Sends a daily absence notification to the parent via WhatsApp/ Email
- ***Branch 2*** (30% - 69\%): Sends a friendly allowance update warning the parent that 30% the allowed threshold has been reached (X days left).
- ***Branch 3*** (70% - 89% Used): Sends an official warning to the parent via WhatsApp and simultaneously alerts the Academic Counselor/Class Teacher via Email/Slack.
- ***Branch 4*** (90% Used): Dispatches a critical meeting notice to the parent and creates a Google Calendar invitation for an urgent Admin-Parent conference - Plus, updates the student's status flag in Google Sheets to "Locked / Under Admin Review".

Workflow 2: Asynchronous Excuse Processing

(Triggers only when a parent replies on Email)



The Workflow:

Node 1: Webhook Trigger: *Receives the message from the parent (WhatsApp, Email, or Web Form)*

Node 2: IF Node: *Checks if the incoming payload contains an attached image/document.*

- TRUE: Passes binary to Node 3 (OCR Engine).
- FALSE: Bypasses OCR and passes directly to Node 4 (Data Aggregator).

Node 3: OCR Engine Node: Converts handwritten or printed doctor notes into clean text.

Node 4: Data Aggregator & Retrieval: *Collects all context streams into a single JSON packet.*

- Google Sheets → returns `student_name`, `total_absences`, `allowed_limit`
- Vector Store (RAG): Passes document type and absence duration to the vector store → returns exact handbook clauses (e.g., *"Medical notes for 1-3 days require clinic signature and must be submitted within 48h"*).

- Output Packet:

```
"parent_message": "{{ $json.message_text }}",
"ocr_data": "{{ $json.ocr_text }}",
"student_profile": { "name": "Omar Ali", "used_absences": 4, "max_allowed": 10 },
"retrieved_policy": "{{ $json.policy_text }}"
```

Node 5: LLM Evaluation Node: *Evaluates the aggregated data against policy and generates a decision.*

- Output Packet:

```
"recommendation": "APPROVED", // "APPROVED", "REJECTED", or "NO_ACTION_REQUIRED"
"confidence_score": 0.95,
"policy_reason": "Valid clinic note provided for 3 days of flu, adhering to Section 4.1.",
"drafted_parent_reply": "Dear Parent, Omar's medical note for Oct 12-14 has been approved. He has 6 allowed absence days remaining."
```

Node 6: Human-in-the-Loop (HITL) Admin Review Node: *Presents the decision to an administrator for a 1-click confirmation before updating database records.*

Node 7A: **Approval Branch (Executes when Admin clicks** `Approve`**)**

1. Google Sheets / DB Update: Updates attendance status to "Excused" and recalculates `remaining_days = max_allowed - updated_unexcused`.
2. WhatsApp Sender Node: Dispatches `drafted_parent_reply` to the parent's phone number.

Node 7B: **Rejection Branch (Executes when Admin clicks** `Reject`**)**

1. Google Sheets / DB Update: Keeps attendance status as "Unexcused" and logs the rejection note in the administrative ledger.
2. WhatsApp Sender Node: Sends a clarification request to the parent (e.g., *"Your medical note could not be verified because the doctor's signature was missing. Please re-submit."*).

8. AI Feature and Why Rules Are Not Enough

- **Where AI Is Used:** A unified OCR + RAG + LLM engine extracts visual document text, fetches relevant handbook policies, and compares parent messages alongside student records against school rules to generate a single decision, confidence score, and pre-drafted reply.

Why Rules Fail (and AI Is Needed):

- **Unstructured Inputs:** Standard IF/THEN code cannot parse messy handwritten notes, blurry photos, or informal WhatsApp text messages.
- **Rigid Logic:** Hardcoding every policy edge case creates fragile, bloated code; updating rules with RAG requires only uploading a new policy PDF rather than rewriting logic.
- **Fraud Risk:** Code only detects if an attachment exists, whereas AI verifies whether the upload is an actual medical slip rather than an unrelated image or blank page.
- **Robotic Messaging:** Rules rely on static templates, while the LLM generates personalized, context-aware draft replies for admin review.

9. What Stays Human? Risks and Escalation

Human Touchpoints:	• **Final Excuse Approval:** Administrators retain complete authority over marking absences as "Excused" or "Unexcused" by confirming or overriding the LLM's suggested assessment. • **Outreach Authorization:** All auto-drafted parent communications (approvals, rejections, or clarification requests) require a 1-click admin sign-off before dispatching via WhatsApp or Email. • **Policy Dispute Resolution:** Any edge cases, parent appeals, or complex medical situations are handled directly by administrative staff and academic counselors. • **Intervention in Critical Cases:** When a student reaches 90%$ of their total allowed absence limit (Level 3), automated processing halts and routes the profile to a human counselor for intervention and meeting scheduling.
Risks & Escalation Path:	• **OCR Extraction Errors:** Blurry images, low resolution, or unreadable doctor handwriting may lead to incorrect date or name extraction. • **LLM Hallucination & Misinterpretation**: The model could misread ambiguous parent phrasing or generate incorrect confidence scores. • **Document Fraud**: Sophisticated fake or altered medical notes might pass initial AI checks.

10. Available Tools / Sample Data / Access

- **Google Sheets:** Mock Google Sheet containing student IDs, class rosters, historical absence counts, allowed limits, and parent phone/email profiles
- **Vector Store (Chroma):** Attendance policies differ by school, grade, or student type. Using Vector DB, the system evaluates compliance against actual official policy, grounding its decision without hallucinating.
- **OCR (Optical Character Recognition):** Parents will upload photos or PDFs of physical doctors' notes, hospital stamped papers, or handwritten leave requests, an OCR Node converts the document into structured text.
- **OpenAI API / Gemini API:** Powering the LLM Evaluation Node for structured JSON output and drafting parent replies.
- **Gmail or Whatsapp API:** Triggering outreach notices, warning alerts, and receiving parent replies.

11. Two Normal Tests + One Unclear / Failure Test

Test Scenario	Input Data Description	Expected Behavior / Route
Normal Test 1 (Happy Path A)	• A student is absent for the first time, with 0 previous absences and a maximum allowed limit of 10 days.	• The workflow records the absence, calculates 9 remaining days, and sends a normal absence notification to the parent.

Normal Test 2 (Happy Path B)	A student is absent and has 8 previous absences, with a maximum allowed limit of 10 days.	The workflow calculates 1 remaining day, identifies that 90% of the allowed absence limit has been used, sends a critical warning to the parent, and alerts the school administrator for review.
Failure / Edge Test	● A student is marked absent, but the previous absence count is missing or invalid in Google Sheets.	● The workflow detects the missing or invalid data, stops the calculation, and alerts the school administrator for manual review. No warning level is assigned until the data is corrected.

12. How Will You Measure Improvement?

Metric	How It Will Be Measured
1-Time Saved	● Compare the time required to record absences, calculate remaining days, and notify parents before and after automation.
2- Calculation Accuracy	● Verify that the system correctly calculates the remaining absence days out of the 10-day limit.
3- Notification Speed	● Measure the time between recording a student's absence and sending the parent notification.
4- Error Rate	● Track missing data, incorrect calculations, and failed workflow executions.
5- Successful Processing Rate	● Measure the percentage of attendance records processed correctly by the workflow.